Nagur Shareef Shaik

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Summary

Applied Scientist with 6.5+ years developing multimodal deep-learning and foundation-model systems for real-world healthcare, from oculomics-based disease screening to automated clinical report generation, with research in attention mechanisms, multimodal fusion, and longitudinal reporting under missing modalities. Published 25 peer-reviewed works (1K+ citations, H-index 13) at venues including AAAI, NeurIPS, ICASSP, ICIP, and MICCAI. As part of UST's AI Center of Excellence, I deliver high-impact AI across Healthcare (primary), Retail, and Finance, spanning claims and payment integrity, risk adjustment, and clinical coding, shipping production RAG and agentic systems with an AI-first (AIDLC) delivery approach.

Education

Georgia State University — Atlanta, GA
Doctor of Philosophy in Computer Science | GPA: 4.23/4.30 — *Aug 2023 – Dec 2026*

Vignan's Foundation for Science, Technology & Research University — Guntur, AP
Bachelors of Technology (Honours) in Computer Science & Engineering | GPA 3.92/4.0 — *Jun. 2016 – May 2020*

Experience

Georgia State University — Aug 2025 – Present
Graduate Research Assistant | TReNDS Center — *Atlanta, GA*

- Proposed a **probabilistic multimodal oculomics framework** with HSIC-constrained latent factorization and a Self-Critiquing Mixture-of-Experts, achieving SOTA on 7 systemic & ocular diseases from mobile fundus images.
- Designed Dist-RRG, a longitudinal multi-modal radiology report generation framework using **distributional temporal modeling**, **Fisher–Rao geometry**, and **MoE-based uncertainty-aware fusion** to preserve diagnostic ambiguity and improve clinical reliability across variable-length patient histories.
- Developed a **probabilistic multi-modal framework** for medical report generation capable of handling missing modalities, leveraging structured latent representations to improve diagnostic accuracy and robustness.

UST Global Inc. — July 2024 – Present
Machine Learning Engineer II | AI-CoE — *(Remote) Aliso Viejo, CA*

- Modernized a payer's **claims adjudication** system by re-engineering extracts and reporting, raising **payment accuracy** and audit-readiness while cutting manual reconciliation and downstream claim reprocessing.
- Built **PolicyIQ**, an **agentic payment-integrity** system that turns **CMS/payer policy** into **executable, SME-approved claim edits** with a full audit trail, cutting policy-to-edit turnaround from weeks to minutes.
- Built **RiskIQ** for **ICD-10 coding & risk adjustment**: surfaces **value-bearing** charts and benchmarks **SLM** ICD-10 mapping **deterministically** on **MIMIC-IV** for auditable, high-throughput autonomous coding.
- Architected and led **Call Analytics Agents**, an **Agentic AI** system that mines member calls to lift **First-Call Resolution**, deflect repeat contacts, and surface quality & compliance insights for a major payer.
- Architected **BSA DataMap** (agentic profiling → **BigQuery** on GCP, saving **2 FTEs/$250K**) and an **ETL Test Agent** that auto-generates and runs ETL test cases and **SQL**, validated by **LLM-as-a-Judge** in the loop.
- Delivered a **RAG-powered RFI/RFP response engine** on Azure with source-linked, traceable, compliance-ready answers, cutting proposal authoring time **60%**.

Georgia State University — Sep 2023 – May 2025
Graduate Research Assistant | TReNDS Center — *Atlanta, GA*

- Customized **Multi-modal LLMs** for automated report generation from medical images and text using **PyTorch**, achieving a **13.4% higher** BLEU4 than VisionGPT, and reducing inference time to **1.6 sec** per image.
- Developed image classification models using **TensorFlow/Keras**, for diagnosing chronic diseases like retinopathy, schizophrenia, breast cancer, and colon cancer from respective medical images, reducing **5%–7%** false negatives.

Carelon Global Solutions — Sep 2022 – Aug 2023
Software Engineer III | Elevance Health — *Hyderabad, TS*

- Developed **REST APIs** for **CO**mpensation **IN**centive **S**ystem, a **Microservices** based application, to validate, compute, and expedite incentive payments, achieving a **10%** reduction in processing time & enhancing scalability.
- Developed **Python** and **SQL** data clean-up scripts to resolve commission payment inconsistencies in health insurance policies, preventing over **$500K** in overpayments and streamlining processing time.
- Resolved critical production issues in commissions calculation flow, ensuring accurate agent advance computation, saving **$1.5 million** in historical commission overpayments. Received **Go Above Impact Award** for this work.

Tata Consultancy Services Aug 2020 – Sep 2022
Systems Engineer | Analytics & Insights Business Unit *Hyderabad, TS*

- Designed a Bug Root Cause Prediction System based on logs, by implementing **Attention LSTM** model in **Azure ML Studio**, cutting the debugging efforts and saving **3 full-time equivalents** annually.
- Implemented an **Azure DevOps** Model Deployment pipeline, reducing deployment time **from 2 to 1.2 hours** and increasing system availability by **25%**. Recognized as **Star Performer** of the team for this significant work.
- Built a custom SonarQube plugin for static code analysis, integrated into pre-build pipelines, reducing errors and vulnerabilities by **15%**, saving **6 hours** of manual review time per week, and adhering to coding standards.

Skills

Languages: Python, Java, SQL
Technologies: LangGraph, LangChain, Flask, FastAPI, Streamlit, Tableau, Spring Boot, REST APIs, Microservices
Agentic Tools: Claude Code, Codex, Cursor, Windsurf, GitHub Copilot
Libraries: PyTorch, torchtune, TensorFlow, Keras, Scikit-Learn, OpenCV, NLTK, NumPy, Pandas, Matplotlib
Cloud & MLOps: MLflow, AWS (EC2, Lambda, SageMaker, Bedrock AgentCore), Azure (AI Services, DevOps), GCP (Agentspace, Gemini Enterprise), CI/CD, Git, Docker
Research Interests: Deep learning, Multi-modal Learning, Attention Networks, Large Language Models, AgenticAI
Certifications: Claude Certified Architect Professional, Azure AI Fundamentals, Deep Learning Specialization, Python for Everybody

Projects

AURA | Github | *Python, FastAPI, React/TypeScript, Vision Transformers, Google Gemini, ChromaDB* Nov 2025

- Built **AURA**, a multi-agent healthcare ecosystem with a master **intent router** that delegates multi-intent queries to specialist agents for diagnosis, CBT-based wellbeing, triage, nutrition, and health visualization.
- Engineered **IRIS**, a **Vision Transformer** oculomics engine predicting systemic risk (diabetes, hypertension, cardiovascular) from a mobile retinal image, and **COMPASS**, an **Agentic RAG** insurance guide, benchmarked on mBRSET, MIMIC-CXR, DeepEyeNet & ROCO.

NL2Dash | Github | *Python, NL2SQL, LLMs, FastAPI, React* Sep 2025

- Built a **self-serve analytics** tool that generates **person-specific dashboards** directly from raw data via natural-language queries, with **NL2SQL** as the core engine translating NLQ into governed SQL and auto-composing visualizations, eliminating BI-team dependency for ad-hoc reporting.

AgentKit | Github | *Python, LangGraph, Agentic Frameworks, Observability* Aug 2025

- Developed an **agent-orchestration library** atop open-source agentic frameworks that adds a production **harness**, pluggable **memory**, **observability**, and **evaluation**, standardizing agent development and cutting time-to-production for new agents across teams.

PRISM | Github | *Python, FastAPI, React, AI Agents (LangGraph),LLMs, Vector DB* Oct 2025

- Built a **next-gen AI-powered data intelligence platform** integrating **NL2SQL**, automated ETL, one-click ML pipelines, and glass-box agentic workflows to turn diverse enterprise data into actionable insights within minutes.
- Engineered a unified intelligence stack, **Data Insights**, **RAG Factory**, **Predictive Modelling**, enabling no-code auto-config RAG, cross-database analytics, and secure LLM reasoning with full auditability.

InsuCompass | Github | Demo | *Python, LangGraph, Streamlit, Groq, Google Gemini, ChromaDB* May 2025

- Developed an **Agentic RAG** health insurance advisor featuring a self-learning **Search Agent** that autonomously retrieves, verifies, and ingests new knowledge (CMS.gov, VA.gov), with multi-turn, personalized plan guidance.
- Orchestrated a multi-agent workflow using **LangGraph** to deliver context-aware insurance advice, leveraging the **Groq API** powered **LlaMA 3**, **Google Gemini Pro**, and **Flash** models, integrated with real-time web search.

ScholarPulse | Github | Demo | *Python, LangChain, Streamlit, ChromaDB, SentenceTransformers* April 2025

- Architected an AI-powered research assistant based on **Advanced RAG** to analyze complex academic papers, delivering tailored question-answering, summarization, and code generation for users with diverse backgrounds.
- Engineered a **LangChain** workflow with advanced prompt engineering for **LlaMA-4 Maverick** insights and **Qwen-2.5** code generation, in a **multi-stage RAG pipeline**, cutting research comprehension time by **50%**.

Retinal Health Diagnostics | Github | Demo | *Python, TensorFlow, FastAPI, Streamlit* May 2024

- Developed an AI system with a novel attention mechanism that leverages global context to learn localized lesion-specific features for diagnosing chronic retinal diseases.
- Achieved state-of-the-art performance with accuracies of 97.5% for cataracts, 85.6% for diabetic retinopathy, and 94.6% for macular edema, highlighting strong clinical relevance.

Doctoral Dissertation

From Perception to Clinical Reasoning: Structured & Disentangled Latent Modeling for Robust Multi-Modal Medical AI | Advisor: Dong Hye Ye — Aug. 2023 – Aug. 2026

- Proposed a **sequential multi-modal pipeline** that resolves the structural mismatch between standard neural-network assumptions and clinical data geometry via **modality-aware attention**, **asymmetric cross-modal fusion**, and **probabilistic latent disentanglement**, validated across **7 clinical domains**, **13 benchmarks**, and **8 novel architectures** (proposal defended Apr 2026).
- Designed modality-aware attention and **vision–text–genomics fusion** across retinal, endoscopic, and neuroimaging data, achieving **+2–7%** over generic attention and **SOTA** report generation and classification (**0.231** BLEU-4; **86.05%** accuracy).
- Introduced **probabilistic latent disentanglement** (VLVAE, WAE, Mixture-of-Experts) to prevent catastrophic failure under missing modalities, class imbalance, and representation collapse.
- Developed **Dist-RRG** (Temporal Disentangled Alignment) for **longitudinal** report generation, modeling disease progression as **distributional change on statistical manifolds** via **Fisher–Rao geodesics** and **Moment-Matched Distributional Aggregation**, achieving **SOTA** clinical accuracy (**+7.4% CE-F1**) on Longitudinal MIMIC-CXR; under review at **NeurIPS 2026**.
- Developed **STRIDE**, an **uncertainty-aware longitudinal clinical reasoning** framework using **distributional change modeling**, **Fisher–Rao geometric attention**, and **recursive natural-gradient reasoning**, outperforming strong **longitudinal report-generation baselines** while improving clinical accuracy and uncertainty calibration; under review at **AAAI 2027**.

Master's Thesis

Attentive Multi-modal Learning for Medical Image Analysis | Advisor: Dong Hye Ye — Aug. 2023 – May. 2025

- Advanced **multi-modal LLMs** and **attention networks** to integrate medical images, clinical text, and genetics, improving diagnostic accuracy for schizophrenia and automating medical report generation.
- Focused on scalable, explainable AI systems for deployment in resource-constrained environments, contributing to cutting-edge research in **Generative AI** and **multi-modal learning**.

Research Publications

- Advanced research at the intersection of **machine learning, computer vision, and NLP**, focusing on scalable attention networks, multi-modal learning, transformers and vision-language models for medical AI, resulting in **1K+ citations**, **13 H-index**, and impactful contributions to diagnostic decision-support systems (Google Scholar).
- Published **27** peer-reviewed papers (**9** conference, **18** journal, and workshop) with **oral presentations** at **AAAI 2026**, **EMBC 2025 & 2026**, **MLSP 2026**, **ICIP 2024**, and **ISBI 2024**, plus works at **ICASSP 2025**, **ISBI 2025**, and **NeurIPS 2025** (GenAI4Health); **Dist-RRG** under review at **NeurIPS 2026**.
- Collaborated with **Stanford AI Lab** on **AxisSQL** (**inference-time scaling**), a **Text-to-SQL** reasoning framework benchmarked on **BIRD**, achieving **#12 on the BIRD leaderboard** at the time of submission (Leaderboard).
- Actively contributed to the research community as a reviewer for **25+ journals** and major conferences (AAAI, MICCAI, NeurIPS, EACL, ISBI), evaluating **60+ papers** across AI/ML, biomedical imaging, and computer vision.

Accomplishments

- Recipient of the **Graduate Research Award 2026** from Georgia State University, recognizing outstanding PhD research contributions and peer-reviewed publications.
- Winner of the **D3CODE Hackathon 2025** (US Region & Global) by UST Global Inc., outperforming top international teams from the UK, Mexico, Malaysia, and India.
- Awarded the **NeurIPS 2025 GenAI4Health Workshop Travel Grant** and the **IEEE SPS Travel Grant** for **ICIP 2024**; recipient of a **100% graduate scholarship** at Georgia State University in recognition of academic excellence and research potential.
- Honored with multiple corporate excellence awards, including the **Go Above – Impact Award** at Elevance Health (Carelon Global Solutions) and the **Star Performer Award** at Tata Consultancy Services for critical code-base analysis and impactful project contributions.
- Recognized for outstanding academic achievements with the **Chairman's Gold Medal**, **Best Outgoing Student of CSE**, and **4-year Academic Excellence Awards** at *VFSTR University*, along with multiple wins in national-level technical expos and paper presentation competitions.